hic lc3 literature excerpt

Synthetic training content. Not a real Holcim trial report.

LC3 literature excerpt (synthetic teaching digest)

> This file is a **teaching digest** of publicly discussed LC3 themes. It is not a journal paper and not an HIC result. Never present numbers here as Loire Plant or HIC-LC3-24 data.

What published LC3 work is generally about

Limestone calcined clay cement (LC3) replaces a large share of clinker with calcined kaolinitic clay and limestone. Public papers often discuss:

- Clinker factors in the region of 0.50 for ?LC3-50? type blends
- 28-day strengths that can approach a 42.5 class when clay quality and fineness are controlled
- Early-age (1-day) strengths that are often lower than CEM I at the same water/binder
- Water demand that can be higher than CEM I, depending on clay

Illustrative literature ranges (not HIC)

These ranges are **teaching placeholders** for ?what papers often say.? They are not measurements from HIC-LC3-24.

- LC3-50 28-day mortar strength: often discussed around 35 MPa to 50 MPa depending on clay and protocol
- 1-day strength: often reported lower than a CEM I control in the same study
- Clinker factor: often near 0.50 for LC3-50 narratives

If a notebook answer copies 35 MPa to 50 MPa into an HIC results table, that answer is wrong. HIC numbers come only from the trial brief or the plant-handover note.

How to use this file in the review

- Use it to explain why a plant manager worries about 1-day strength
- Use it to explain why water demand came up
- Do not use it to fill a missing HIC cell